

RC Low-Pass Filter PCB Design (KiCad)

Cal Poly Strawberry Center – OAS Intern Onboarding

Khiem Vu

1. Project Overview

This project documents the design and implementation of a simple RC low-pass filter PCB developed as part of the Operator Aid System (OAS) onboarding process at the Cal Poly Strawberry Center.

The circuit filters frequencies higher than the cutoff frequency from an input signal, allowing only lower frequency signals to pass to the output. This project serves as an introduction to using KiCad for schematic capture, PCB layout, design validation, and manufacturing file generation.

2. Circuit Design

The circuit is an RC low-pass filter composed of a resistor (R1) and capacitor (C1). The input signal is applied across the resistor, and the output is taken at the node between the resistor and capacitor.

Using **R1 = 10kΩ** and **C1 = 0.1μF**, the **cutoff frequency** is approximately **159 Hz**. This configuration attenuates frequencies above 159 Hz while passing lower frequencies.

The cutoff frequency is defined as: $f_c = \frac{1}{2\pi RC}$

These values were selected because they are standard, readily available, and produce a cutoff frequency suitable for observing filtering behavior in a lab environment.

3. Schematic

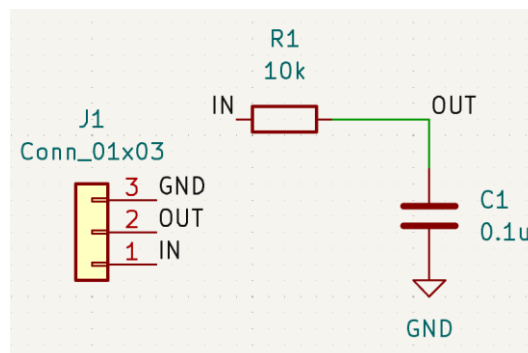


Fig. 1: RC circuit schematic in KiCad

- **Signal Flow**
 - Input (IN) → R1 → [Filtered output node (OUT)] → C1 → GND.
- **Connector Usage**
 - IN (input), OUT (filtered output), GND (reference).

4. Bill of Materials

Designator	Value	Footprint	Description	Mount
R1	10k Ω	R_0805_2012Metric	Resistor	SMT
C1	0.1 μ F	C_0805_2012Metric	Capacitor	SMT
J1	-	PinHeader_1x03_P2.54mm_Vertical	1x3 Connector	THT

5. PCB Layout

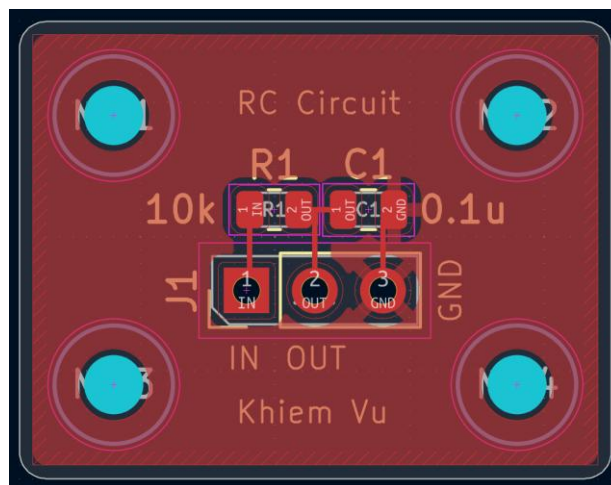


Fig. 2: RC circuit in KiCad PCB Editor

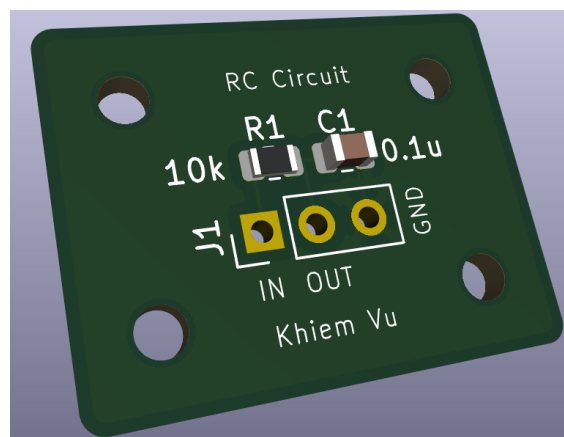


Fig. 3: RC circuit in 3D (KiCad)

- **Layout Decisions**
 - PCB layout is compact and optimized for short, direct signal paths and clarity of signal flow between components.
 - As shown in Fig. 2, components are arranged to reflect the signal flow (IN→R1→OUT→C1→GND)
- **Design Considerations**
 - Short traces reduce parasitic inductance and resistance, which helps minimize signal degradation and parasitic effects.
 - Copper GND plane
 - Provides a low impedance return path for the signals that are in the circuit.
 - Widespread copper on the GND plane can act as an EMI shield.
 - Allows heat generated to be spread out across a wider area.
 - Cheaper to manufacture (fewer copper cuts).
 - Clean routing improves signal integrity.
 - Silkscreen labels provide visual clarity about PCB components.
- **Other Specifications**
 - This board measures approximately 22.0mm x 17.0mm.
 - 0.5mm clearance between copper pour and board boundary.

6. Layer Stackup

- **2-Layer PCB**
 - F.Cu (Top Layer)
 - All components
 - All signal traces
 - GND copper pour
 - B.Cu (Bottom Layer)
 - No active routing (present for manufacturability)
 - Two layers are sufficient due to low component count and simple routing requirements.

7. Footprint Choices

- **0805 SMD Components (R1, C1)**
 - Standard and widely available.
 - Appropriate size for manual soldering.
- **Through-Hole Header (J1)**
 - Mechanically robust for repeated connections.
 - Easier for prototyping and testing with lab equipment.
 - Standard 2.54mm header pins.
- **M2 Mounting Holes**
 - Enables stable integration into fixtures or enclosures.

8. Design Rules and Validation

- **KiCad DRC passed with zero errors**
- Default KiCad trace width: 0.2mm
 - o Sufficient for this low-frequency low-current circuit.
- Copper-to-edge clearance: 0.5mm
 - o Standard manufacturing requirement.

9. Manufacturing Files

- Gerber files, drill files, and Bill of Materials (BOM) were generated using KiCad's PCBWay plugin.

10. Lessons Learned

- This project provided me with a practical introduction to the PCB design workflow in KiCad, from schematic capture to manufacturable outputs. I learned to think beyond the circuit schematic, considering other factors such as component placement, strategic routing, footprint choices, and grounding strategy. This demonstrated how schematic-level design decisions can directly impact physical layout considerations. Without prior experience in KiCad, this project also helped me get more familiar with the software and its various features.